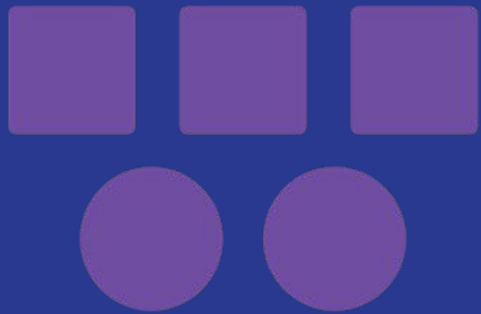




WeGo
Public Transit

Effect of TSP strategy on WeGo bus performance

Nashville Software School- 'fast coding bosses'
Sarah Mulloy, Abigail Ezell, Anitha Doddala
January 2026

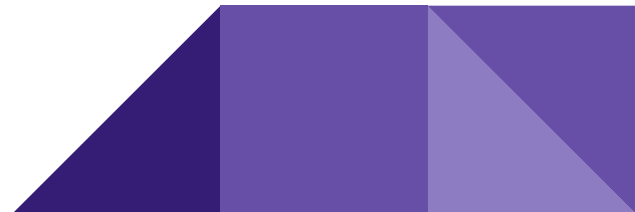
Key Terms/Abbreviations

WeGo: public transit system serving the Nashville area

Transit Signal Priority (TSP): a technology that manipulates traffic signals to manage traffic flow more efficiently

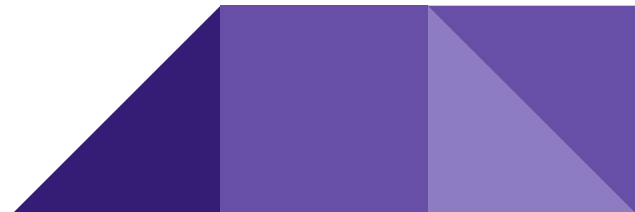
Adherence: the difference between a bus's actual departure time and its scheduled departure time

Headway: the difference between the scheduled time for a particular stop and the scheduled time for the previous bus on that same stop



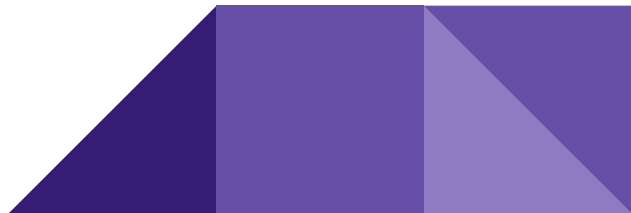
Project Purpose

- Main research question: **Is Traffic Signal Priority (TSP) effective?**
 - Does TSP reduce bus lateness?
 - Does TSP reduce unacceptable headway?
- We analyzed WeGo Public Transit data for trips on Route 50, Charlotte Pike between 1-1-2023 and 5-12-2025
- TSP was enabled and disabled for different portions of the time period



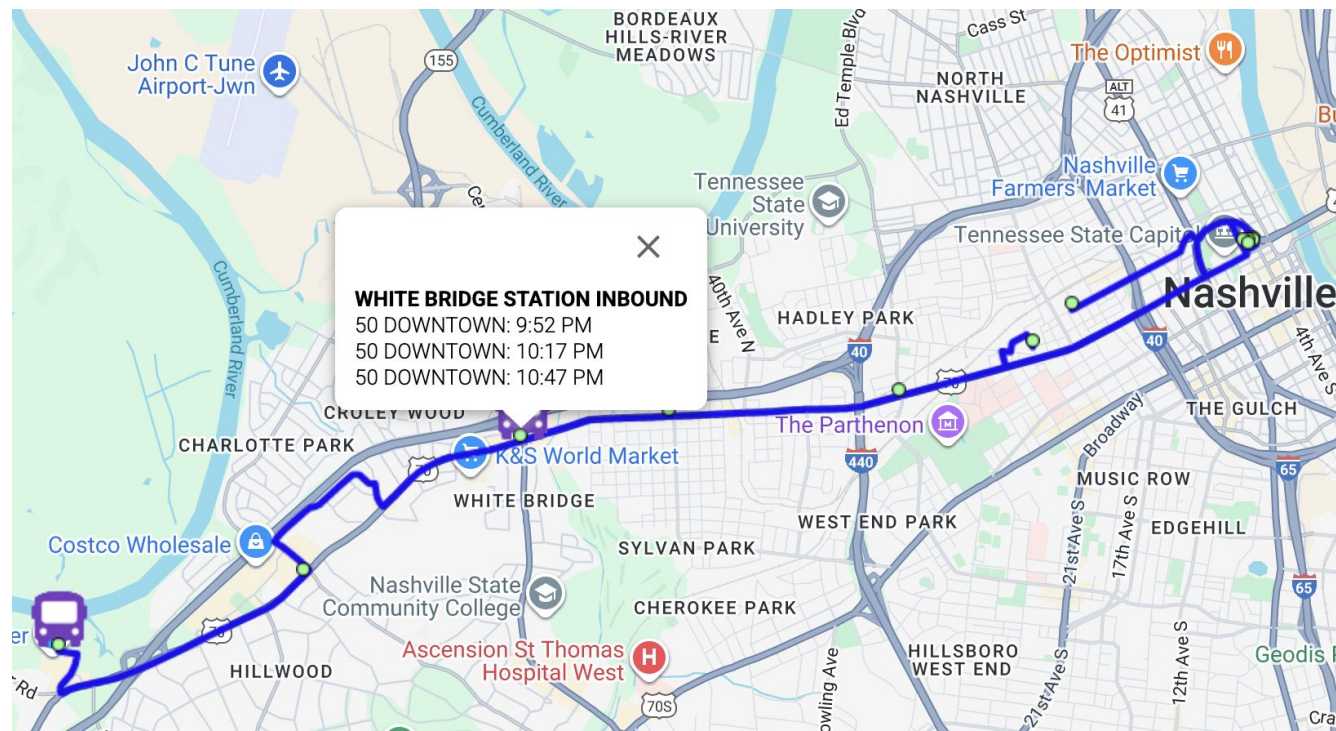
Data Strategy & Cleaning

- Dropping any rows that have missing values for either ACTUAL_ARRIVAL_TIME or ACTUAL_DEPARTURE_TIME
- Cleaning up SCHEDULED_TIME, ACTUAL_ARRIVAL_TIME, and ACTUAL_DEPARTURE_TIME columns, removing "1900-01-01" from entries that have it
- Dropping the row with a strange value for ACTUAL_ARRIVAL_TIME (-** :03:34)
- After data cleaning, we retained 97.06% of the original data



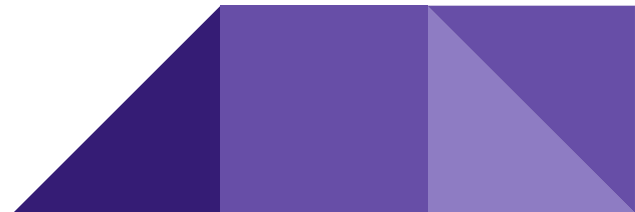
Data Preparation

- We used data from WeGo route #50, Charlotte Pike



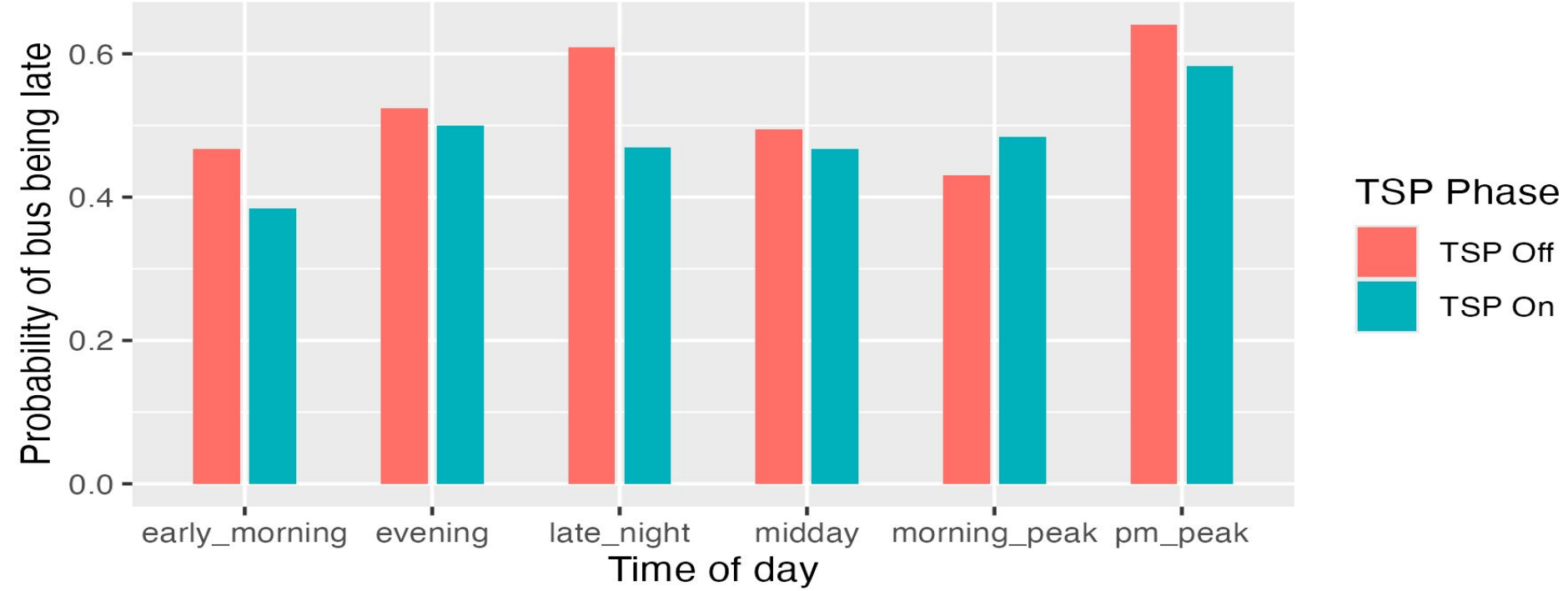
Data Preparation

- We created a new TSP column based on TIME_POINT_ABBR and ACTUAL_ARRIVAL_TIME
 - Also broke out additional columns for TSP type
- In addition, we created columns for time of day, day of week, and whether school is in session or not based on SCHEDULED_TIME

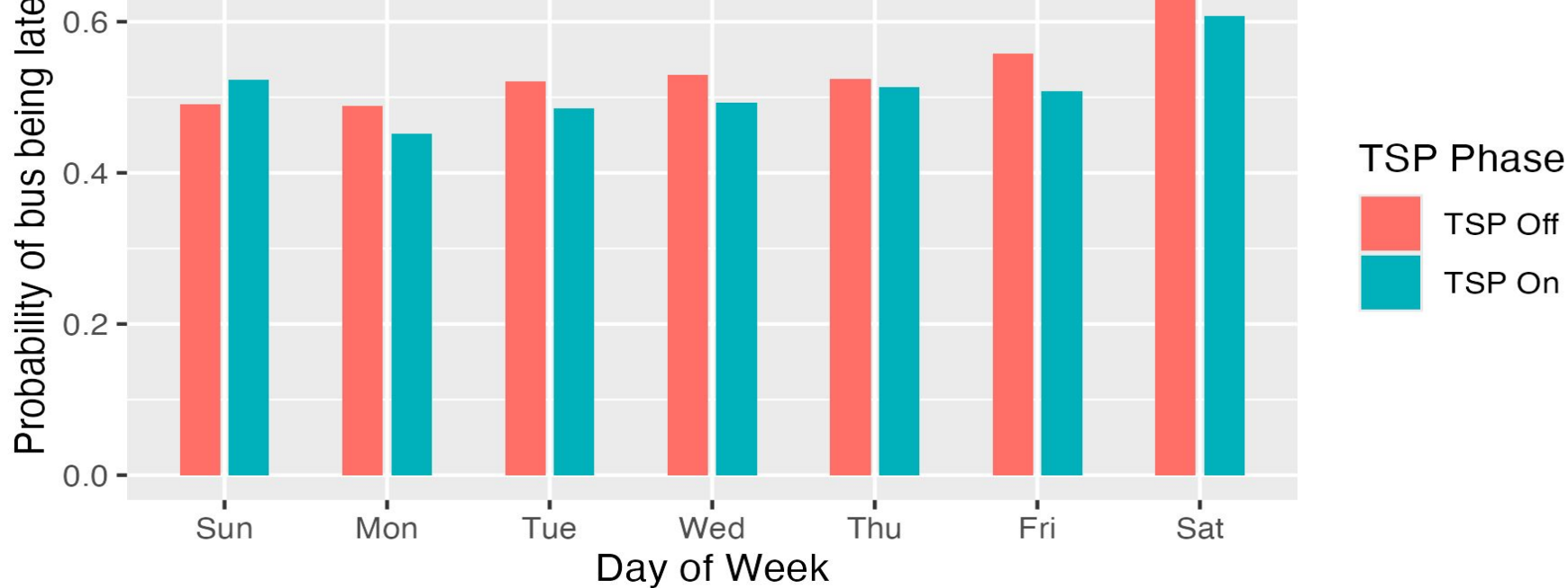




Does Transit Signal Priority (TSP)
reduce the probability that a bus
leaves a stop late?

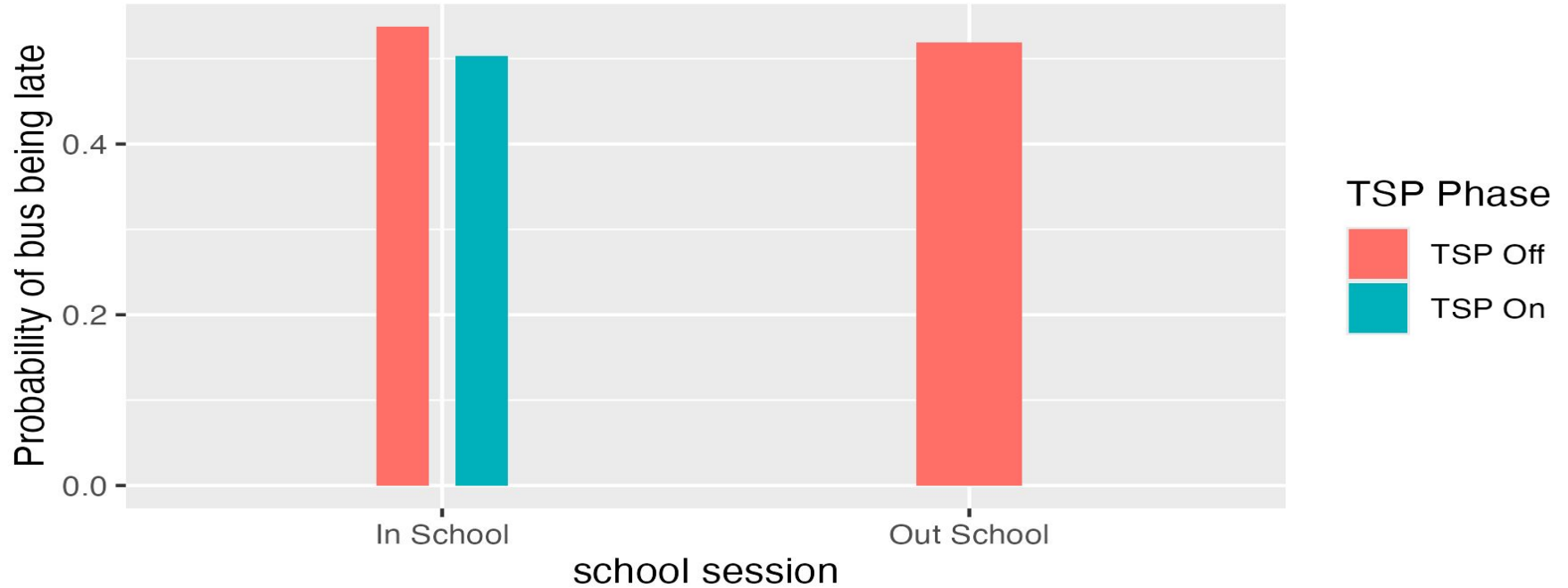


- Buses are less likely to be late when TSP is active, especially during peak times.



- Buses are less likely to be late when TSP is active, across most of the days in a week.
- Sunday shows slightly higher lateness under TSP, it might be due to lower traffic.

The probability of bus lateness by school session and TSP Phase



- TSP was not active during out school sessions in this data set.
- During in school sessions activating TSP reduces the Probability of bus being late.

The Model

	Estimate	Pr(> z)
(Intercept)	-0.144640314	4.329950e-01
tsp_phasetsp_on	-0.145193788	8.755597e-06
time_of_dayevening	0.236507201	2.055601e-01
time_of_daylate_night	0.445807897	1.865794e-02
time_of_daymidday	0.113289405	5.391742e-01
time_of_daymorning_peak	-0.075552105	6.840623e-01
time_of_daypm_peak	0.712671576	1.161105e-04
day_of_week.L	0.493965337	1.008074e-51
day_of_week.Q	0.197593356	1.089886e-09
day_of_week.C	0.152446756	1.507769e-07
day_of_week^4	0.090218205	5.823567e-04
day_of_week^5	-0.028119955	2.591309e-01
day_of_week^6	-0.002625153	9.149426e-01
school_sessionOut School	-0.085135390	2.469075e-03
TIME_POINT_ABBRCH46	0.003564531	8.752145e-01
TIME_POINT_ABBRWHBG	0.172232894	2.658173e-10

Buses operating with active TSP have approximately 13.5% lower odds of bus being late compared to periods when TSP is off, controlling for time of day, day of week, school session and stop location.



What effect does TSP have on bus headway?

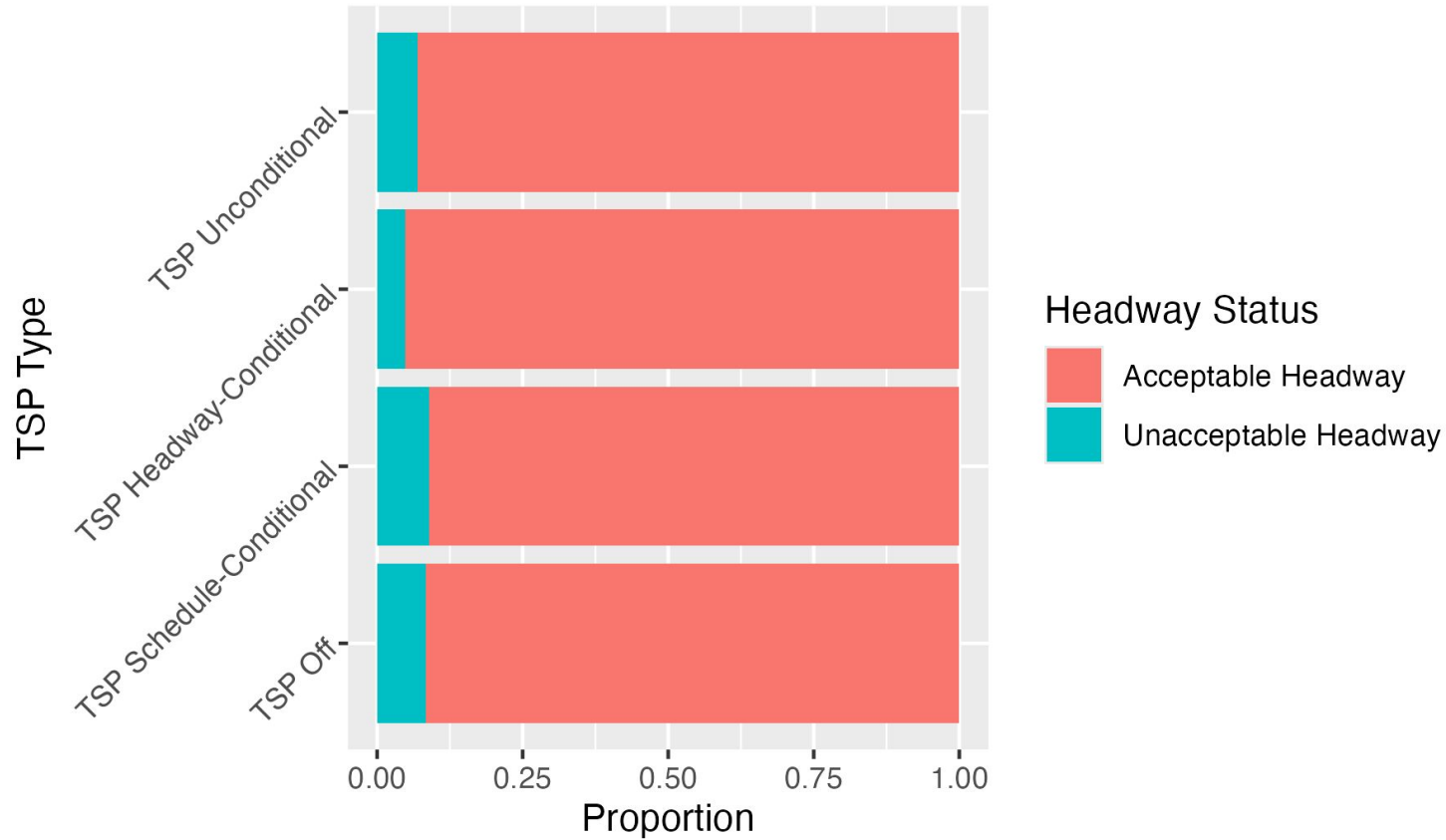
Effect of TSP on Headway

- We also examined the effect of TSP on headway (amount of time between a bus and the prior bus at the same stop).
- We want to deviate no more than 50% from scheduled headway in either direction.



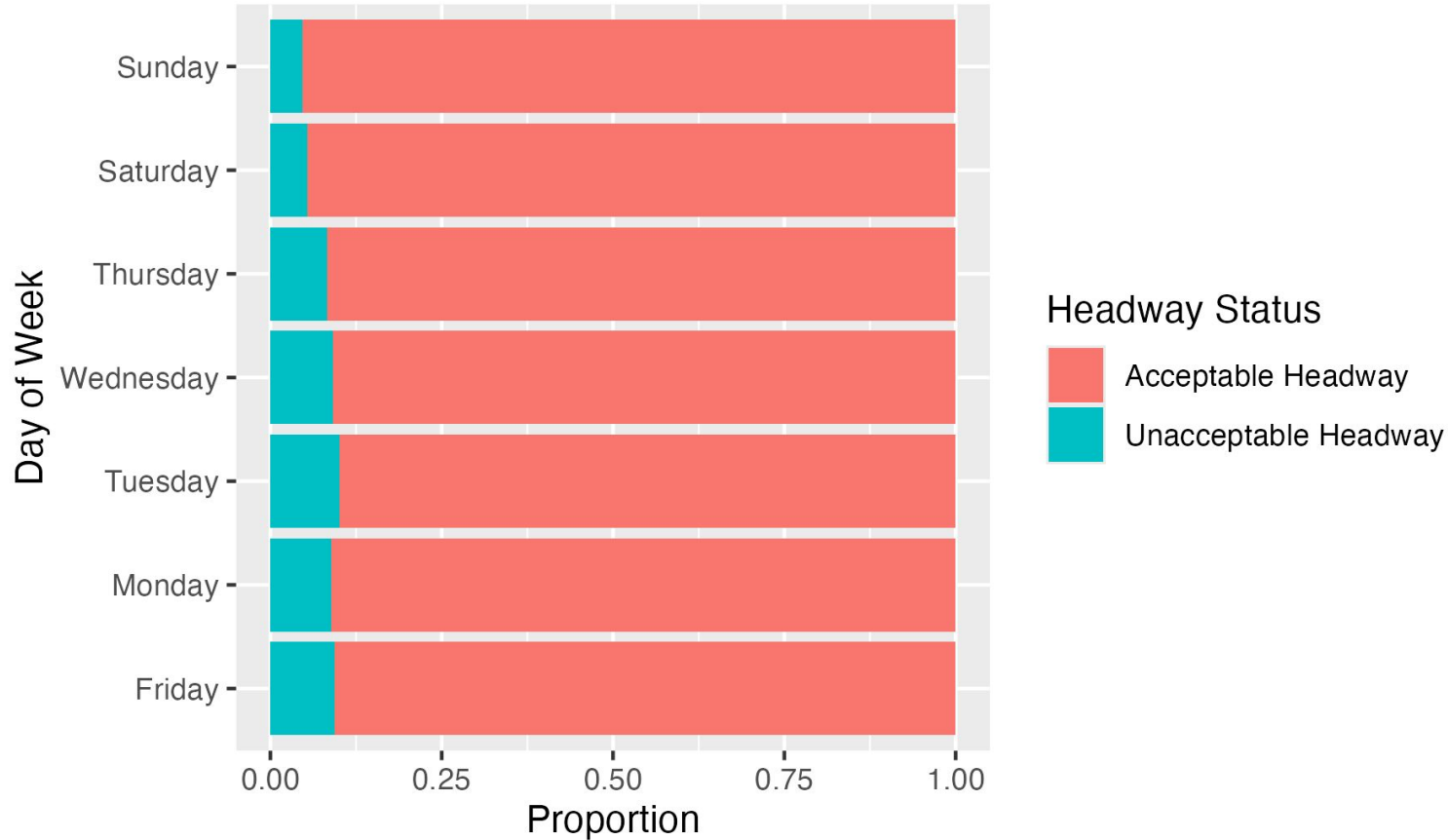
Frequency of Excessive Headway by TSP Status

For WeGo Bus Rides, 1/1/2023 - 5/12/2025



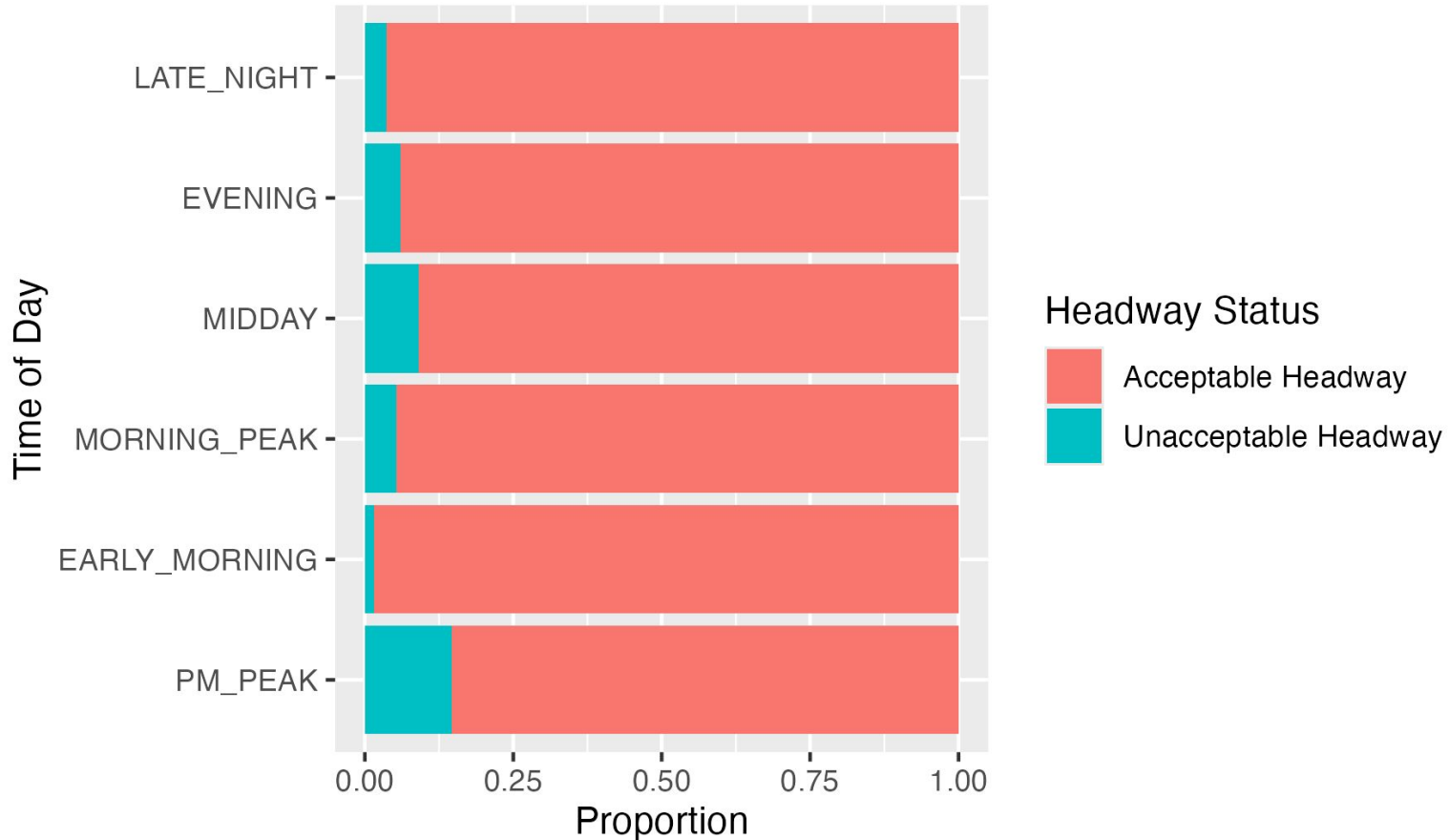
Frequency of Excessive Headway by Day of Week

For WeGo Bus Rides, 1/1/2023 - 5/12/2025



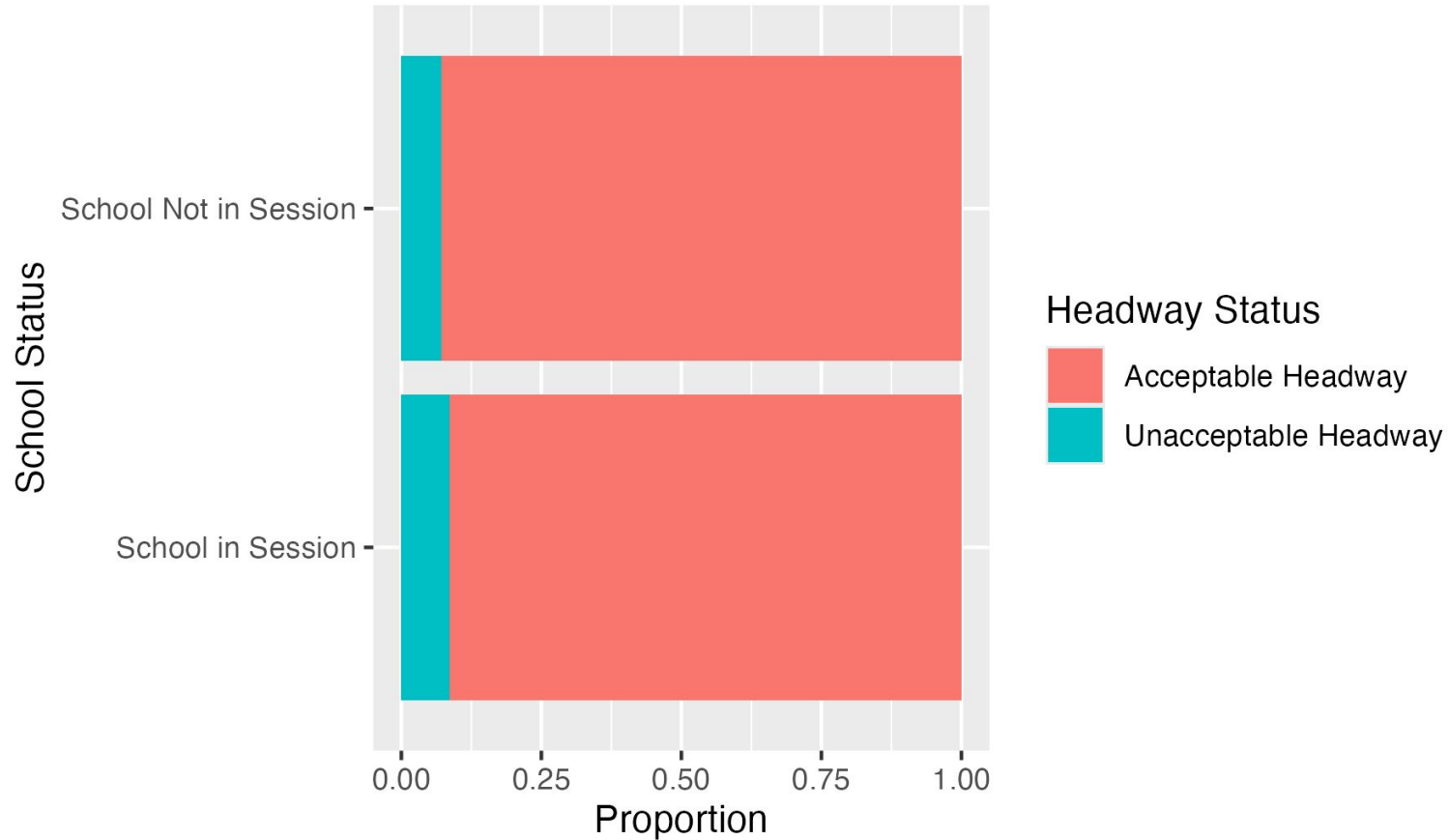
Frequency of Excessive Headway by Time of Day

For WeGo Bus Rides, 1/1/2023 - 5/12/2025



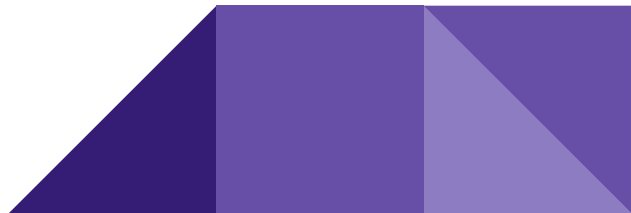
Frequency of Excessive Headway by In-School Status

For WeGo Bus Rides, 1/1/2023 - 5/12/2025



Logistic Regression Model

- We fit various logistic regression models based on different combinations of these variables and chose the model with the lowest AIC
- The model we chose incorporated TSP Type, time of day, day of week, and whether or not school was in session



The Model

Term <chr>	Coefficient <dbl>	P-Value <dbl>
(Intercept)	-1.525943033	0.0000
TSP_TypeTSP Schedule-Conditional	0.050548121	0.5363
TSP_TypeTSP Unconditional	-0.147264495	0.1604
TSP_TypeTSP Headway-Conditional	-0.721262574	0.5804
TIME_OF_DAYEARLY_MORNING	-3.369183015	0.0000
TIME_OF_DAYMORNING_PEAK	-1.254795121	0.0000
TIME_OF_DAYMIDDAY	-0.639530874	0.0000
TIME_OF_DAYEVENING	-1.076140953	0.0000
TIME_OF_DAYLATE_NIGHT	-1.590199177	0.0000
DAY_OF_WEEKMonday	-0.059295961	0.1784

We applied the Bonferroni correction to the p-values in our model since we had many explanatory variables.

Results

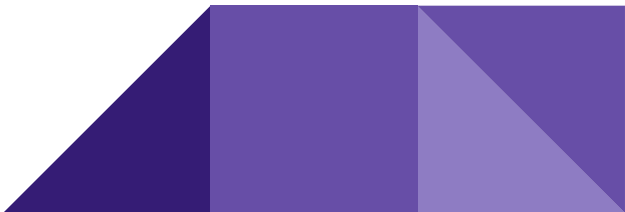
After controlling for day of week, time of day, and whether school is in session...

Unconditional TSP reduces the odds of a bus having unacceptable headway by about **13.7%.***

Schedule-conditional TSP *increases* the odds of a bus having unacceptable headway by about **5.2%.***

Headway-conditional TSP reduces the odds of a bus having unacceptable headway by about **51.4%.***

* Not statistically significant

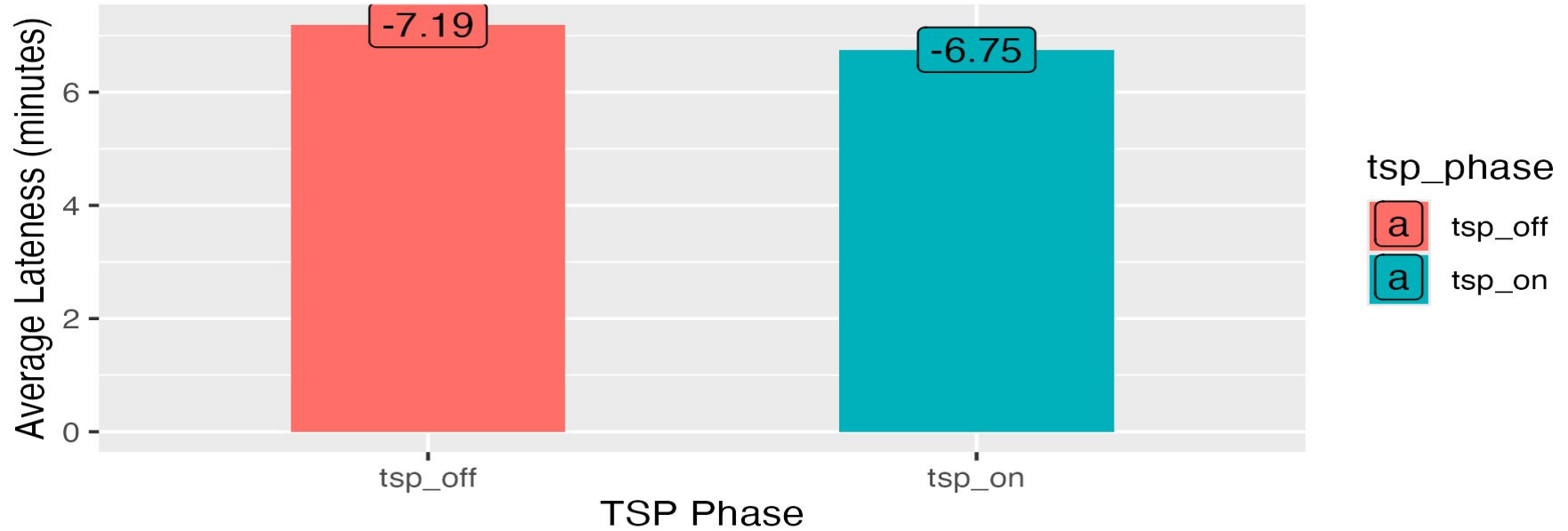
A decorative graphic in the bottom right corner consisting of several overlapping purple geometric shapes, including triangles and rectangles, in various shades of purple.



How much does the TSP reduce the average bus delay time?

Effect of TSP phase on Bus Lateness

higher values = more late



Linear Regression Model

	Estimate	Pr(> t)
(Intercept)	-6.346440728	1.879290e-26
tsp_phasetsp_on	0.497652492	1.962471e-06
time_of_dayevening	-0.723979155	2.303499e-01
time_of_daylate_night	-1.393855591	2.269437e-02
time_of_daymidday	-0.151736816	7.990399e-01
time_of_daymorning_peak	0.349196475	5.604029e-01
time_of_daypm_peak	-1.956664396	1.050539e-03
day_of_week.L	-1.584901936	4.151073e-53
day_of_week.Q	0.309830894	2.526545e-03
day_of_week.C	-0.817780196	7.958575e-19
day_of_week^4	0.077065847	3.577709e-01
day_of_week^5	-0.030890894	6.990533e-01
day_of_week^6	-0.139847355	7.629135e-02
school_sessionOut School	0.158957969	7.742306e-02
TIME_POINT_ABBRCH46	-0.006958309	9.237185e-01
TIME_POINT_ABBRWHBG	-0.569261733	5.903200e-11

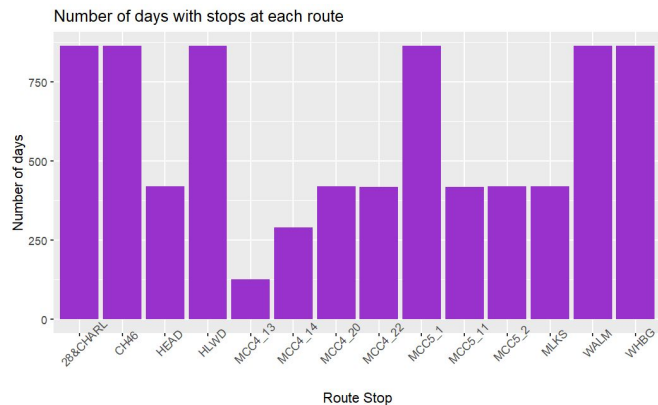
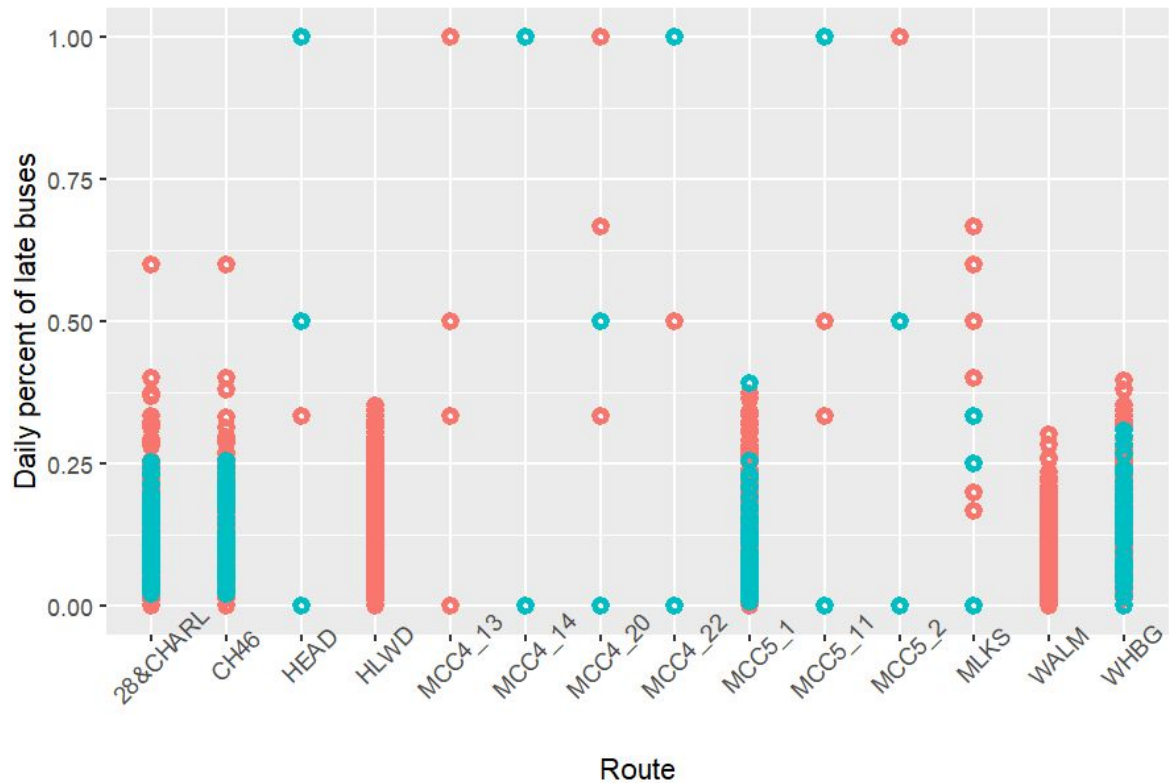
- When TSP is activated, average bus schedule adherence improve by approximately 0.50 minutes (about 30 seconds), indicating reduced lateness.
- The coefficient for TSP activation is statistically significant ($p = 1.96 \times 10^{-6}$), showing that TSP has a significant association with bus schedule adherence after controlling other variables like Time of Day, Day of Week, School Session and stop location.





Does TSP improve performance at
each route stop?

Daily percent of bus late per route using TSP



```
call:
lm(formula = pc_late ~ tsp_on + TIME_POINT_ABBR, data = late_tsp_date_route)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.22397	-0.06129	-0.02487	0.03153	1.00134

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	0.118096	0.005737	20.587	< 2e-16	***
tsp_onon	-0.048872	0.006736	-7.255	4.37e-13	***

TARGET: percent of daily buses that were late

Predictor 1 = TSP on/off
Predictor 2 = Route stop

Results

Regardless of stops, TSP being on reduced percent bus lateness significantly by 0.04% daily



Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	0.118096	0.005737	20.587	< 2e-16	***
tsp_onon	-0.048872	0.006736	-7.255	4.37e-13	***
✕TIME_POINT_ABBRCH46	0.003245	0.008058	0.403	0.687215	
●TIME_POINT_ABBRHEAD	-0.068447	0.010243	-6.683	2.50e-11	***
●TIME_POINT_ABBRHLWD	0.017179	0.008269	2.077	0.037800	*
✕TIME_POINT_ABBRMCC4_13	-0.010096	0.016668	-0.606	0.544747	
●TIME_POINT_ABBRMCC4_14	-0.070563	0.011782	-5.989	2.20e-09	***
●TIME_POINT_ABBRMCC4_20	0.098458	0.010268	9.589	< 2e-16	***
●TIME_POINT_ABBRMCC4_22	0.103529	0.010284	10.067	< 2e-16	***
●TIME_POINT_ABBRMCC5_1	-0.022915	0.008058	-2.844	0.004469	**
●TIME_POINT_ABBRMCC5_11	0.105873	0.010285	10.294	< 2e-16	***
●TIME_POINT_ABBRMCC5_2	-0.059058	0.010276	-5.747	9.38e-09	***
✕TIME_POINT_ABBRMLKS	-0.012852	0.010243	-1.255	0.209590	
●TIME_POINT_ABBRWALM	-0.029844	0.008269	-3.609	0.000309	***
●TIME_POINT_ABBRWHBG	0.024619	0.008031	3.066	0.002179	**

signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1					

TARGET: percent of daily buses that were late

Predictor 1 = TSP on/off
Predictor 2 = Route stop

P-value had bonferroni correction

Results

TSP effect varied based on routes. Some routes increased percent lateness and others reduced percent lateness.

*Linear regression -> effects of route on daily percent lateness regardless of TSP status